CampusFlow – Implementation Plan (v2)

CampusFlow is a **unified college platform** for students, faculty, and administrators. It covers the full spectrum of campus life — academics, administration, communication, and automation. This is a hackathon project, so the focus is on a **polished, functional MVP** with a stunning UI and intelligent automation via **n8n workflows**.

User Review Required

> *[!IMPORTANT] **Technology Stack** – **React.js (Vite)** (frontend) + **Supabase** (backend/auth/DB/storage) + **n8n** (workflow automation). Confirm if this works for you.*

> *[!IMPORTANT] **n8n Deployment** – n8n can run locally via Docker (* `docker run -it --rm -p 5678:5678 n8nio/n8n` *) or as a cloud instance. Let me know your preference.*

> *[!WARNING] **Scope** – This plan now has **14 modules**. For the hackathon demo, I recommend building the first 8-9 modules fully and having the rest as UI shells. Let me know what to prioritize.*

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1. Architecture Overview

```
graph TD
    subgraph Frontend ["React App - Vite"]
        UI[Pages & Components]
        Auth[Auth Context]
        API[API Utility Layer]
    end

    subgraph Backend ["Supabase"]
        SB_AUTH[Auth]
        SB_DB[PostgreSQL]
        SB_STORAGE[Storage]
        SB_REALTIME[Realtime]
    end

    subgraph Automation ["n8n Workflow Engine"]
        WF1["Leave Approval Notifications"]
        WF2["Attendance Alerts"]
        WF3["Event Reminders"]
        WF4["Document Status Updates"]
        WF5["Fee Due Reminders"]
        WF6["Complaint Escalation"]
    end

    UI --> Auth --> SB_AUTH
    UI --> API --> SB_DB
    UI --> API --> SB_STORAGE
    UI --> SB_REALTIME

    SB_DB -- "Webhooks / DB triggers" --> Automation
    Automation -- "Email / SMS / Push" --> UI
```

Layer	Technology	Why
Frontend	React.js 18 (Vite)	Fast HMR, lightweight, pure React
Routing	React Router v6	Client-side routing, nested layouts
Styling	Vanilla CSS + CSS Variables	Full control, premium design
Auth	Supabase Auth	Email/password + OAuth, role-based
Database	Supabase (PostgreSQL)	Relational data, Row-Level Security
Storage	Supabase Storage	Document uploads, profile pics
Realtime	Supabase Realtime	Live notifications, mail
Automation	n8n	Visual workflow builder, webhook triggers, email/SMS sending

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2. User Roles & Permissions

Role	Access
Student	View attendance, request documents/leave, browse events, pay fees, use AI chatbot
Faculty	Mark attendance, approve documents/leave, create events, view analytics
Hostel Warden	Manage rooms, approve hostel leave, handle complaints
Admin	Full access, manage users, configure n8n workflows, view college-wide analytics
Club Lead	Create club events, manage club members, post announcements

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3. Core Modules (14 Total)

🔐 Module 1 – Authentication & Dashboard

- Login / sign-up with role selector
- Role-based dashboard: stats cards, quick actions, notification feed
- Profile management with avatar upload

📋 Module 2 – Attendance System

- **Faculty**: Mark attendance per class/subject/date
- **Student**: View attendance %, subject-wise breakdown
- Calendar heat-map visualization
- **n8n**: Auto-email students when attendance drops below 75%

📄 Module 3 – Document Manager

- Request documents (bonafide, transcripts, ID cards, NOCs, migration certs)
- Approval workflow with status tracking (Pending → Approved → Ready)
- Upload signed documents via Storage
- **n8n**: Notify student via email when document status changes

🏠 Module 4 – Hostel Manager

- Room allocation dashboard with block/floor/room grid
- Complaint / maintenance request system with priority levels
- Room swap requests
- Hostel fee status
- **n8n**: Escalate unresolved complaints after 48 hours

🌿 Module 5 – Leave Manager

- Apply for leave with date range, reason, and attachment
- Multi-level approval (student → faculty → HOD)
- Leave balance tracker (casual, medical, academic)
- **n8n**: Auto-notify approver on new leave request; escalate if pending > 2 days

🎉 Module 6 – Events & Club Management

- Create, browse, and RSVP to events
- Event categories (cultural, technical, sports, academic, workshops)
- Club pages with member lists, posts, and event history
- Calendar view + card grid view
- **n8n**: Send reminder emails 24h before event; auto-generate attendance report post-event

✉️ Module 7 – Internal Mail / Messaging

- Compose, inbox, sent, starred, trash
- Role-based recipient search ("all CS faculty", "all 3rd year students")
- Bulk announcements
- Realtime notification badges via Supabase Realtime

💰 Module 8 – Fee Management

- View fee breakdown (tuition, hostel, exam, library)
- Payment status tracker (paid, pending, overdue)
- Receipt generation (PDF download)
- **n8n**: Send payment reminder 7 days before due date; send overdue alerts

📚 Module 9 – Library Management

- Book catalog with search and filters
- Issue / return tracking
- Fine calculation for overdue books
- Reservation system
- **n8n**: Auto-send return reminder 2 days before due date

📊 Module 10 – Timetable & Schedule

- Weekly timetable view per class/section
- Faculty schedule view
- Room/lab booking for extra classes
- Conflict detection

📝 Module 11 – Exam & Results

- Exam schedule display
- Result publication with GPA/CGPA calculation
- Semester-wise result history
- Grade distribution charts

🤖 Module 12 – AI Campus Assistant (Chatbot)

- Floating chatbot on every page
- Answers FAQs (exam dates, fee deadlines, hostel rules, etc.)
- Powered by a simple rule-based + keyword-matching engine (no external AI API needed for hackathon)
- Quick links to relevant platform pages

📢 Module 13 – Noticeboard & Announcements

- Admin/faculty post notices with category tags
- Pinned / urgent announcements
- Department-wise filtering
- Push-style notification feed on dashboard
- **n8n**: Auto-distribute urgent notices via email to all relevant users

🛡️ Module 14 – Grievance & Feedback System

- Submit anonymous or named grievances
- Category-wise routing (academic, hostel, ragging, infrastructure)
- Status tracking for the submitter
- Admin resolution workflow
- **n8n**: Route grievances to the correct authority automatically; escalate unresolved after 72h

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4. n8n Workflow Summary

#	Workflow Name	Trigger	Action
1	Low Attendance Alert	Supabase webhook on attendance insert	Check if student < 75%, send email
2	Document Status Notifier	Supabase webhook on document update	Email student with new status
3	Leave Request Notifier	New leave_request row	Notify approver via email
4	Leave Escalation	Cron (daily)	Check pending leaves > 2 days, email HOD
5	Event Reminder	Cron (daily)	Send reminder for events happening in 24h
6	Fee Payment Reminder	Cron (weekly)	Email students with upcoming/overdue fees
7	Complaint Escalation	Cron (daily)	Escalate hostel complaints open > 48h
8	Library Return Reminder	Cron (daily)	Email for books due in 2 days
9	Urgent Notice Distributor	Supabase webhook on notice insert	Mass email for urgent notices
10	Grievance Router	Supabase webhook on grievance insert	Route to correct authority + escalate if unresolved

5. Database Schema (Key Tables)

Table	Key Columns
`users`	id, email, name, role, department, year, avatar_url
`attendance`	id, student_id, faculty_id, subject, date, status
`documents`	id, student_id, type, status, remarks, file_url, created_at
`hostel_rooms`	id, room_no, block, floor, capacity, occupants[]
`hostel_complaints`	id, student_id, room_id, category, description, priority, status
`leave_requests`	id, user_id, approver_id, type, start_date, end_date, reason, status
`events`	id, title, description, date, time, venue, category, created_by, rsvp_count
`clubs`	id, name, description, lead_id, logo_url
`club_members`	id, club_id, user_id, role
`mail`	id, sender_id, recipient_id, subject, body, is_read, starred
`fees`	id, student_id, type, amount, due_date, paid, receipt_url
`library_books`	id, title, author, isbn, available_copies
`library_issues`	id, book_id, student_id, issue_date, due_date, return_date, fine
`timetable`	id, class, section, day, time_slot, subject, faculty_id, room
`exams`	id, subject, date, time, venue, type
`results`	id, student_id, exam_id, marks, grade
`notices`	id, title, body, category, is_urgent, pinned, created_by, created_at
`grievances`	id, user_id, category, description, is_anonymous, assigned_to, status
`feedback`	id, user_id, target_type, target_id, rating, comment

6. File Structure

```
campusflow/
├── public/
├── index.html                  # Vite entry HTML
├── src/
│   ├── main.jsx                # React entry point
│   ├── App.jsx                 # Root component + React Router setup
│   ├── pages/                  # Route-level page components
│   │   ├── Login.jsx
│   │   ├── Dashboard.jsx
│   │   ├── Attendance.jsx
```

```
│   │   ├── Documents.jsx
│   │   ├── Hostel.jsx
│   │   ├── Leave.jsx
│   │   ├── Events.jsx
│   │   ├── Clubs.jsx
│   │   ├── Mail.jsx
│   │   ├── Fees.jsx
│   │   ├── Library.jsx
│   │   ├── Timetable.jsx
│   │   ├── Exams.jsx
│   │   ├── Noticeboard.jsx
│   │   └── Grievances.jsx
│   ├── components/
│   │   ├── layout/              # Sidebar, Navbar, AppLayout
│   │   ├── ui/                  # StatsCard, DataTable, Modal, Calendar, Badge
│   │   ├── chatbot/             # AI Campus Assistant widget
│   │   └── charts/              # Attendance charts, grade dist, analytics
│   ├── lib/
│   │   ├── supabase.js          # Supabase client init
│   │   └── utils.js             # Helpers
│   ├── context/
│   │   └── AuthContext.jsx      # Auth provider + protected routes
│   ├── hooks/                   # Custom React hooks
│   │   ├── useAuth.js
│   │   └── useSupabase.js
│   └── styles/
│       ├── globals.css           # Design system tokens
│       └── modules/              # Per-component CSS
├── n8n-workflows/                # Exported n8n workflow JSON files
│   ├── low-attendance-alert.json
│   ├── document-status-notifier.json
│   ├── leave-request-notifier.json
│   ├── fee-reminder.json
│   └── ...
├── .env                          # VITE_SUPABASE_URL, VITE_SUPABASE_ANON_KEY
├── package.json
└── vite.config.js
```

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7. UI / Design Direction

Aspect	Approach
Theme	Dark mode primary, glassmorphism sidebar, vibrant gradients
Typography	*Inter* body, *Outfit* headings (Google Fonts)
Colors	Navy `#0a0f1e` bg, electric blue `#3b82f6` + purple `#8b5cf6` accents
Cards	Frosted glass with subtle border glow
Animations	Page transitions, hover micro-animations, skeleton loaders

Layout	Collapsible sidebar + top navbar, fully responsive
Chatbot	Floating bottom-right bubble, expands into chat panel

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8. Build Phases

Phase	Modules	Priority
Phase 1	Scaffolding, design system, auth, dashboard	🔴 Must have
Phase 2	Attendance + n8n alert workflow	🔴 Must have
Phase 3	Document Manager + Leave Manager + n8n notifications	🔴 Must have
Phase 4	Events & Clubs + Noticeboard	🔴 Must have
Phase 5	Internal Mail + Realtime	🟡 High
Phase 6	Fee Management + n8n reminders	🟡 High
Phase 7	Hostel Manager + Grievance System	🟡 High
Phase 8	Library, Timetable, Exams/Results	🟢 Nice to have
Phase 9	AI Chatbot + final polish + demo prep	🟢 Nice to have

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9. Verification Plan

Automated

- `npm run build` — Vite production build check
- `npm run lint` — code quality via ESLint

Browser Testing

- Navigate each page via React Router, verify rendering + CRUD
- Test role-based protected routes (student / faculty / admin)
- Test n8n webhook triggers manually

Manual

- You provide Supabase credentials (`VITE_SUPABASE_URL` , `VITE_SUPABASE_ANON_KEY`) in `.env`
- End-to-end demo flow testing
- Mobile responsiveness check
- n8n workflow import + trigger verification